Exercise 8 – CART

Introduction to Machine Learning

Hint: Useful libraries

R

```r
library(mlr3verse)
library(rattle)
```

Python

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeRegressor, plot_tree
```

Exercise 1: Splitting criteria

Learning goals

1) Perform split computation with pen and paper
2) Understand why additional splits never increase the training risk

Consider the following dataset:

i	$x^{(i)}$	$y^{(i)}$
1	1.0	1.0
2	2.0	1.0
3	7.0	0.5
4	10.0	10.0
5	20.0	11.0

a) Compute the first split point the CART algorithm would find for the given data set (with pen and paper or in `R`, resp. `Python`). Use mean squared error (MSE) to assess the empirical risk.

b) Explain why performing further splits can never result in a higher overall risk with $L2$ loss as a splitting criterion.

Exercise 2: CART hyperparameters

> Learning goals
>
> Understand the effect of CART hyperparameters

In this exercise, we will have a look at two of the most important CART hyperparameters, i.e., design choices exogenous to training. Both the *minimum number of observations required to attempt a split* (`minsplit` in `rpart`, `min_samples_split` in `scikit-learn`) and the *maximum tree depth* (`maxdepth`, resp. `max_depth`) influence the number of input space partitions the CART will perform.

a) How do you expect the number of splits to affect the model fit and generalization performance?

b) Fit a regression tree learner to the `bike_sharing` data (omitting the `date` feature) for

- maximum tree depth $\in \{2, 4, 8\}$ with minimum split size 2, and
- minimum split size $\in \{5, 1000, 10000\}$ with maximum tree depth 20.

What do you observe?

Hint

R

Use the `bike_sharing` task from `mlr3` with the learner `regr.rpart`.

Python

Read the already preprocessed file bike_sharing_for_py.csv and use `DecisionTreeRegressor`.

c) Which of the two options should we use to control tree appearance?

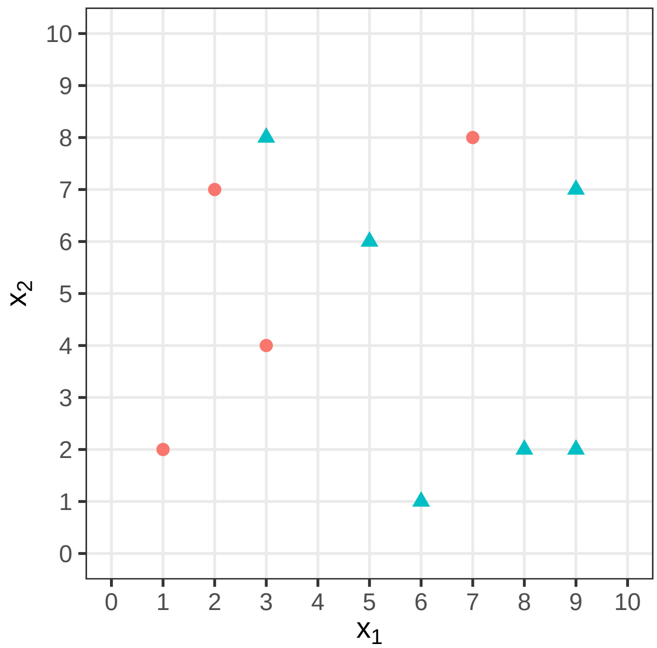
Exercise 3: Drawing CART predictions

The whole exercise is only for lecture group B!

Learning goals

1. Translate between a fitted tree and the corresponding partitioning of the feature space
2. Compute CART predictions by hand

Consider the following classification data with two classes (circles and triangles) and two features x_1 and x_2:



A CART trained on these data performs the following splits:

- first split: $x_1 \leq 4$,
- second split, only for observations with $x_1 > 4$: $x_2 \leq 3$.

a) Draw the partitioning of the feature space induced by the tree into the figure and label the resulting regions R_1, R_2 and R_3.

b) State the predicted class for each region.

Hint

Use the majority of the classes in a region as decision rule.

c) Which class does the tree predict for a new observation $(x_1, x_2) = (7, 2)$?

d) Finally, sketch the tree corresponding to the given splits. Annotate each leaf node with its predicted class.

Exercise 4: Impurity reduction

The whole exercise is only for lecture group A!

Learning goals

1. Derive the optimal constant predictor for regression under L2 loss
2. Develop intuition for use of Brier score in classification trees
3. Reason about distribution and expectations of random variables
4. Handle computations involving expectations

⚠ TLDR;

This exercise is rather involved and requires some knowledge of probability theory. Main take-away (besides training proof-type questions): *In constructing CART with minimal Gini impurity, we minimize the expected rate of misclassification across the training data.*

We take a closer look at the theory behind the splitting criteria for both regression and classification trees.

a) Derive the optimal constant predictor for a node $\mathcal{N}$ when minimizing the empirical risk under $L2$ loss and explain why this is equivalent to minimizing variance impurity.

For classification trees, we now build some intuition for the Brier score / Gini impurity as a splitting criterion by showing that it is equal to the expected MCE of the resulting node.

The fractions of the classes $k = 1, \dots, g$ in node $\mathcal{N}$ of a decision tree are $\pi_1^{(\mathcal{N})}, \dots, \pi_g^{(\mathcal{N})}$, where

$$\pi_k^{(\mathcal{N})} = \frac{1}{|\mathcal{N}|} \sum_{(x^{(i)}, y^{(i)}) \in \mathcal{N}} [y^{(i)} = k].$$

For an expression that holds in expectation over arbitrary data, we need to introduce stochasticity. Assume we replace the (deterministic) classification rule in node $\mathcal{N}$

$$\hat{k} \mid \mathcal{N} = \arg\max_k \pi_k^{(\mathcal{N})}$$

by a randomizing rule

$$\hat{k} \sim \text{Cat}\left(\pi_1^{(\mathcal{N})}, \dots, \pi_g^{(\mathcal{N})}\right),$$

in which we draw the classes from the categorical distribution of their estimated probabilities (i.e., class k is predicted with probability $\pi_k^{(\mathcal{N})}$).

b) Explain the difference between the deterministic and the randomized classification rule.

c) Using the randomized rule, compute the expected MCE in node $\mathcal{N}$ that contains n random training samples. What do you notice?